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Pozzi

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(54) **MULTIMODAL WHEELCHAIR AND METHODS THEREOF**

(56) **References Cited**

U.S. PATENT DOCUMENTS

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 100 days.

5,026,225 A * 6/1991 McIntyre A61G 3/0808 280/808
5,028,065 A * 7/1991 Danecker A61G 5/121 152/323
10,908,045 B2 * 2/2021 Coulter G05D 1/0274
2001/0001031 A1 * 5/2001 Craft A61G 3/0808 410/7

(Continued)

FOREIGN PATENT DOCUMENTS

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CN 102331780 A * 1/2012
WO WO-9112975 A1 * 9/1991 A61G 5/045

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(60) Provisional application No. 63/483,580, filed on Feb. 7, 2023.

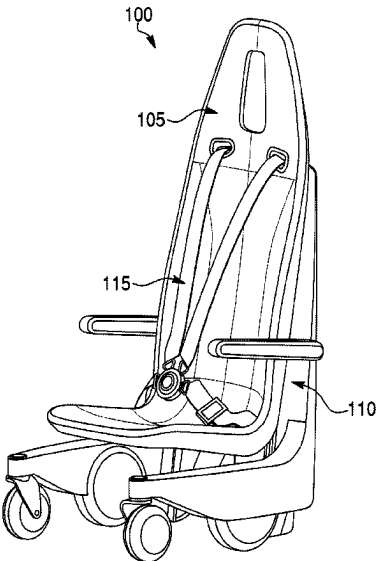
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A61G 3/08 (2006.01)
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A61G 5/12 (2006.01)
G05D 1/43 (2024.01)
G05D 3/12 (2006.01)
G05D 109/10 (2024.01)
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CPC **A61G 5/1059** (2013.01); **A61G 3/0808** (2013.01); **A61G 5/04** (2013.01); **A61G 5/125** (2016.11); **A61G 5/128** (2016.11); **G05D 1/43** (2024.01); **G05D 3/125** (2013.01); **A61G 5/041** (2013.01); **A61G 2203/14** (2013.01); **A61G 2203/20** (2013.01); **A61G 2203/22** (2013.01); **A61G 2203/30** (2013.01); **G05D 2109/10** (2024.01)

(57) **ABSTRACT**
A multimodal wheelchair configured to be securely mounted to an aircraft is disclosed. The multimodal wheelchair may include: a first section, including: a processing system contained within a housing of the first section; a movement controller component positioned on a portion of the first section that is connected to the processing system; a plurality of wheels that are controllable by the movement controller; and a set of tracks recessed within a first side of the first section; and a second section, including: a seat portion located on a first side of the second section, the seat portion having an integrated harness; and a connection component, coupled to a second side of the second section, that attaches the second section to the first section via coupling with the set of tracks of the first section. Other aspects are described and claimed.

(58) **Field of Classification Search**
None
See application file for complete search history.

20 Claims, 19 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

2004/0251656	A1 *	12/2004	Patterson		A61G 7/1053
					280/304.1
2016/0184150	A1 *	6/2016	Kennedy		A61G 5/125
					180/6.7
2019/0282431	A1 *	9/2019	Moore		A61G 5/04
2024/0016673	A1 *	1/2024	Pozzi		A61G 3/0808

* cited by examiner

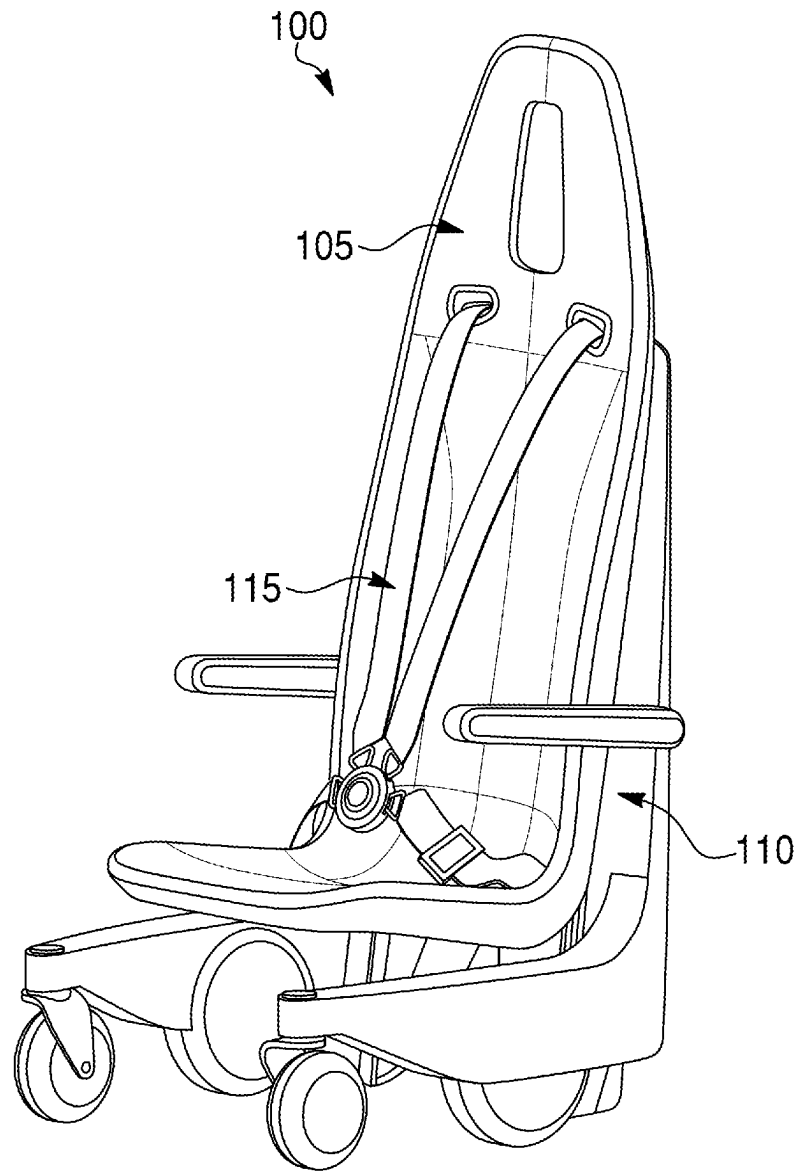


FIG. 1

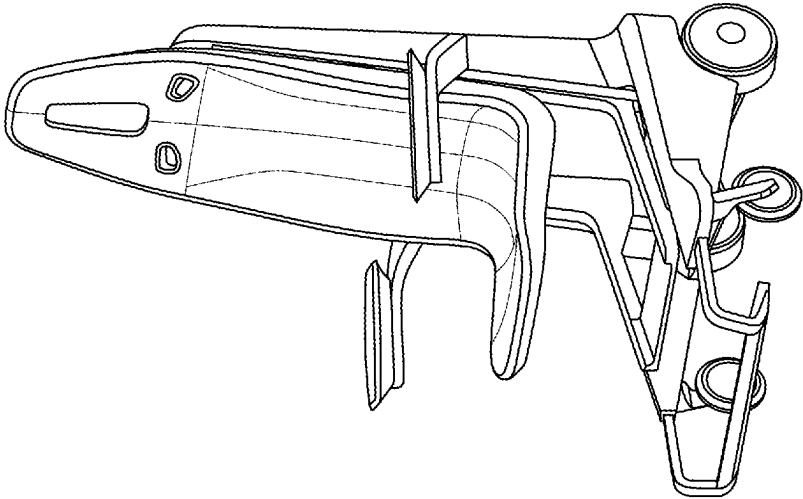


FIG. 2B

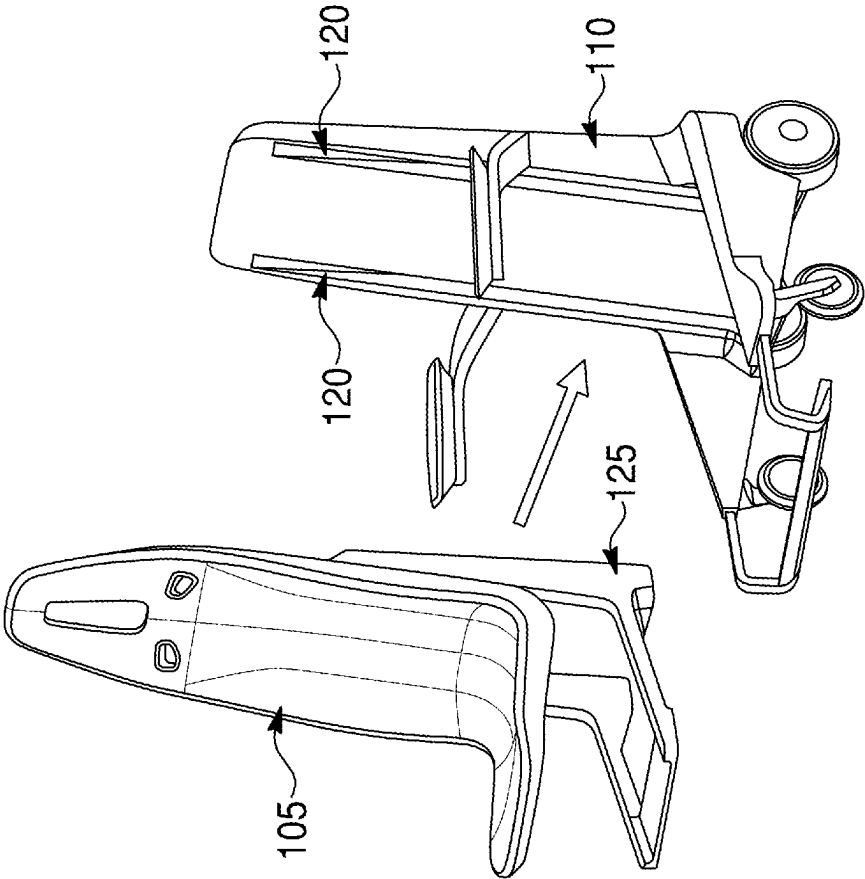
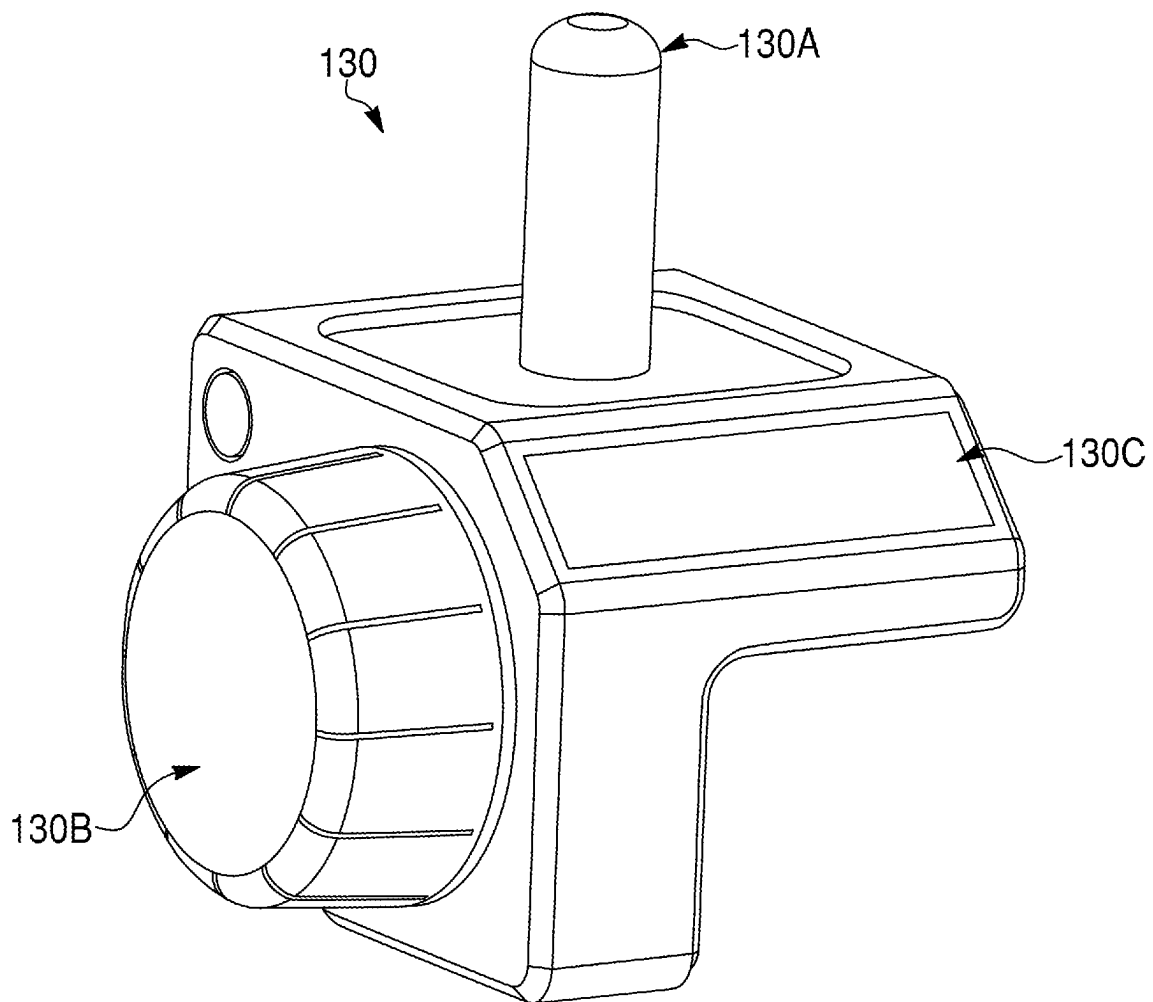


FIG. 2A

**FIG. 3**

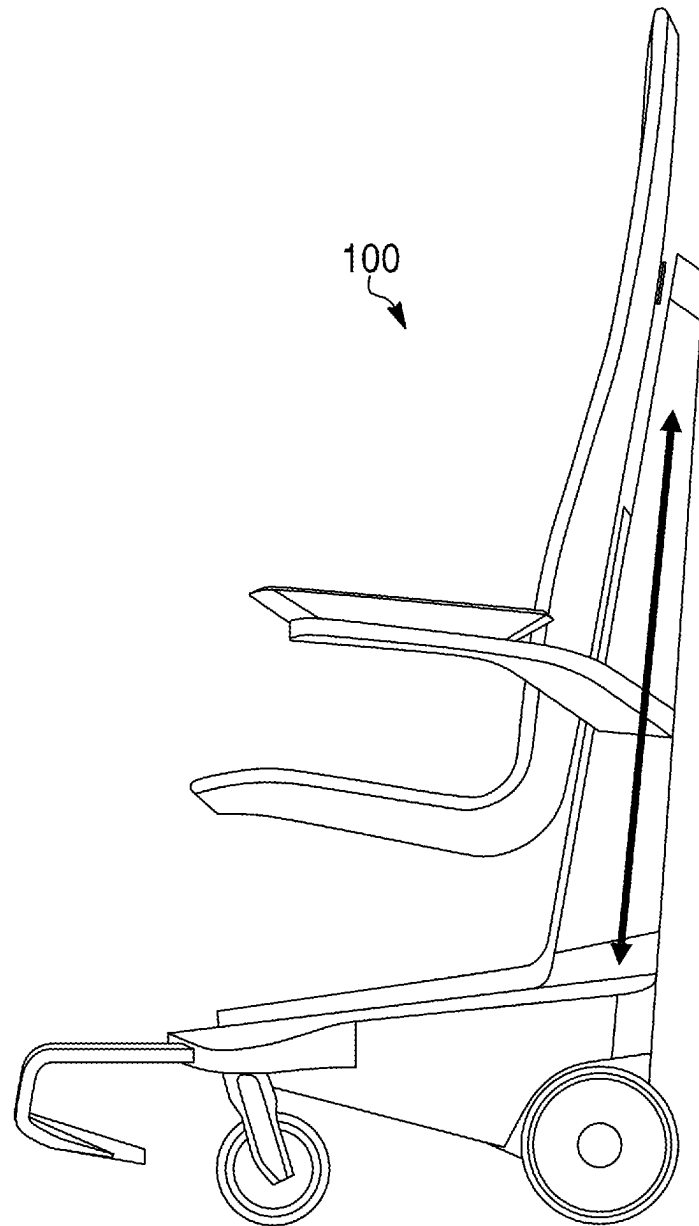


FIG. 4

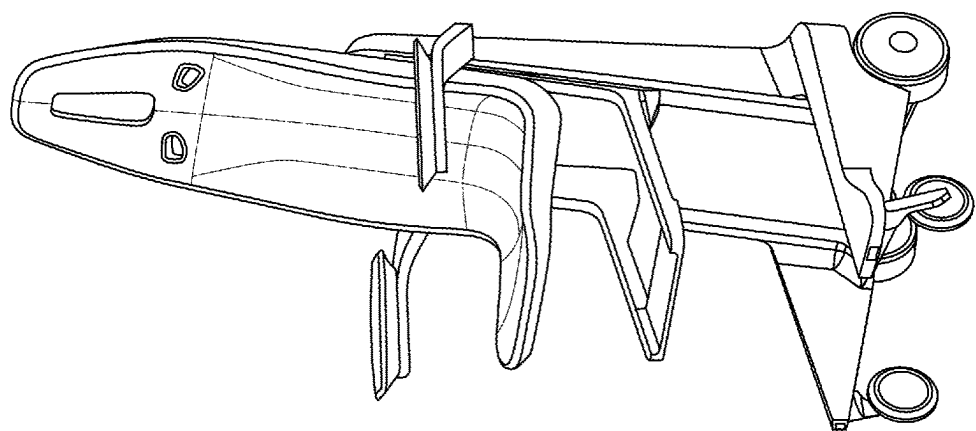


FIG. 5C

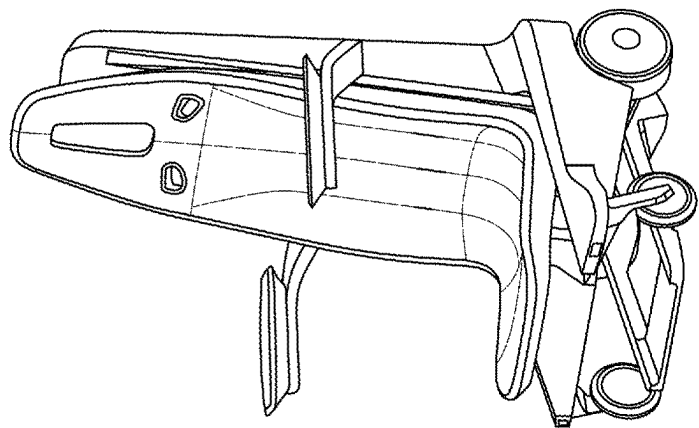


FIG. 5B

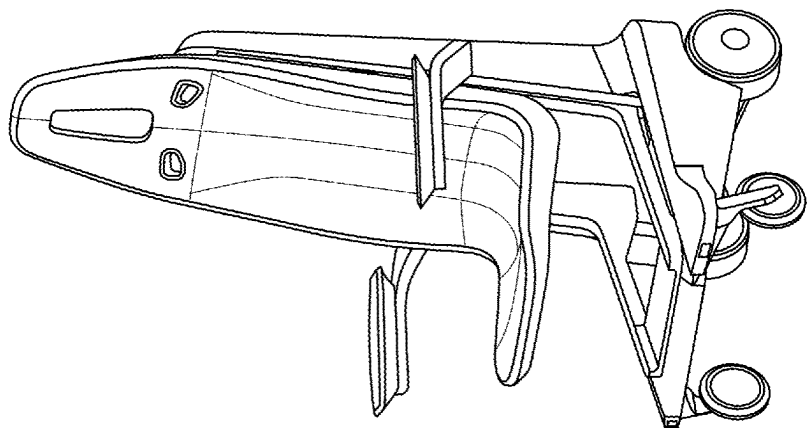


FIG. 5A

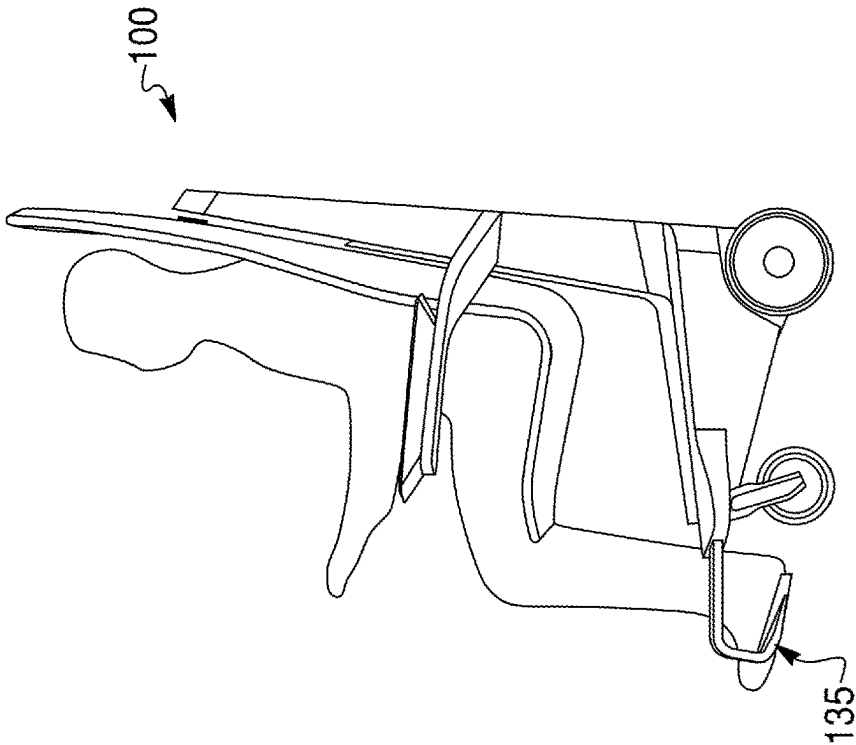


FIG. 6A

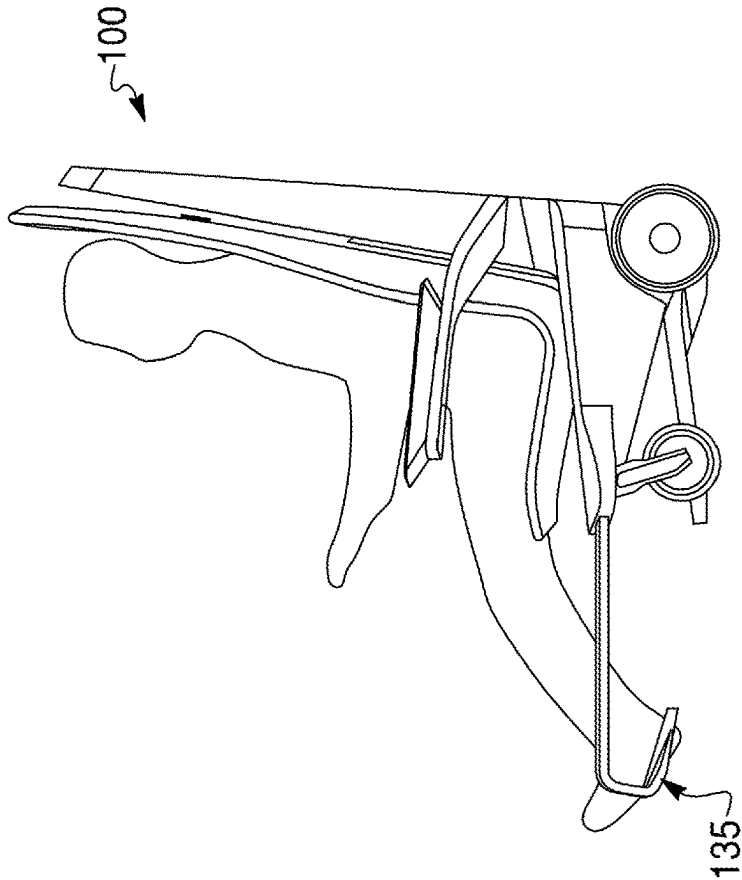


FIG. 6B

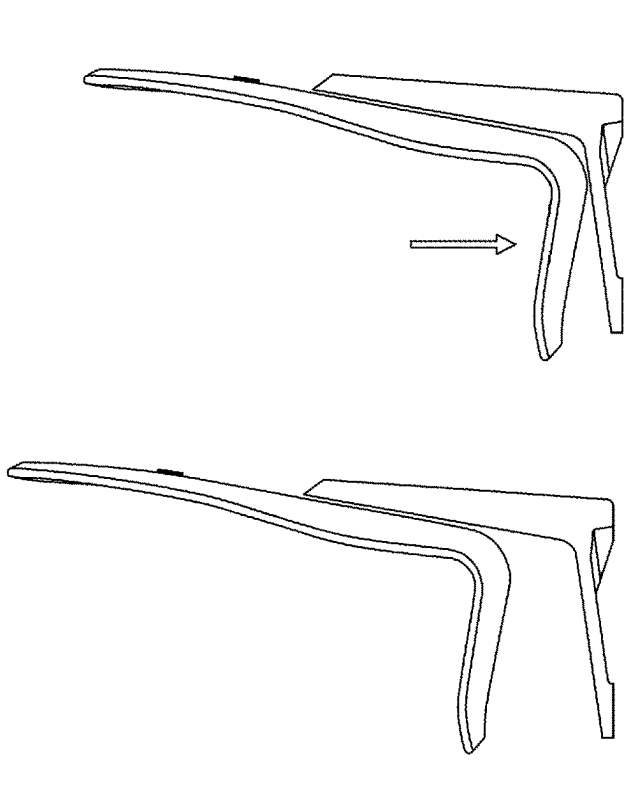


FIG. 7B

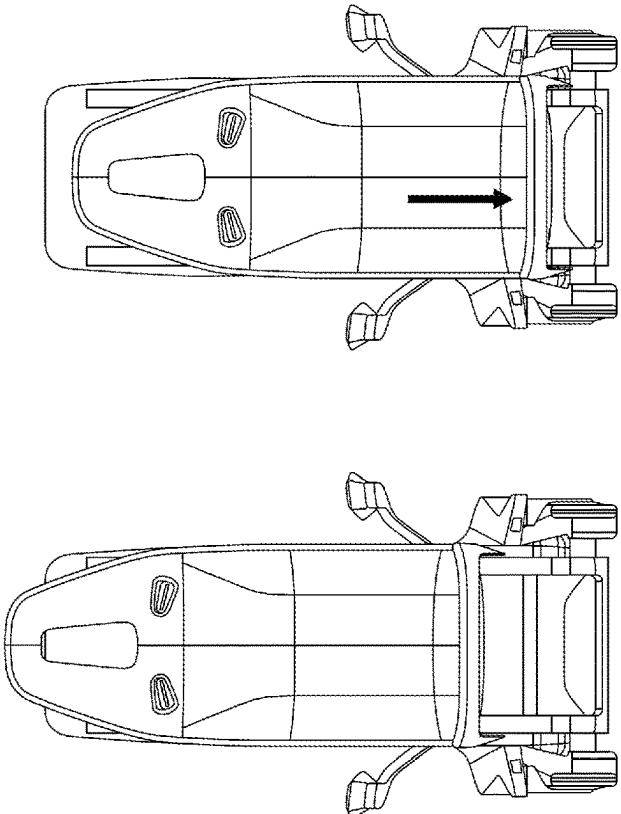
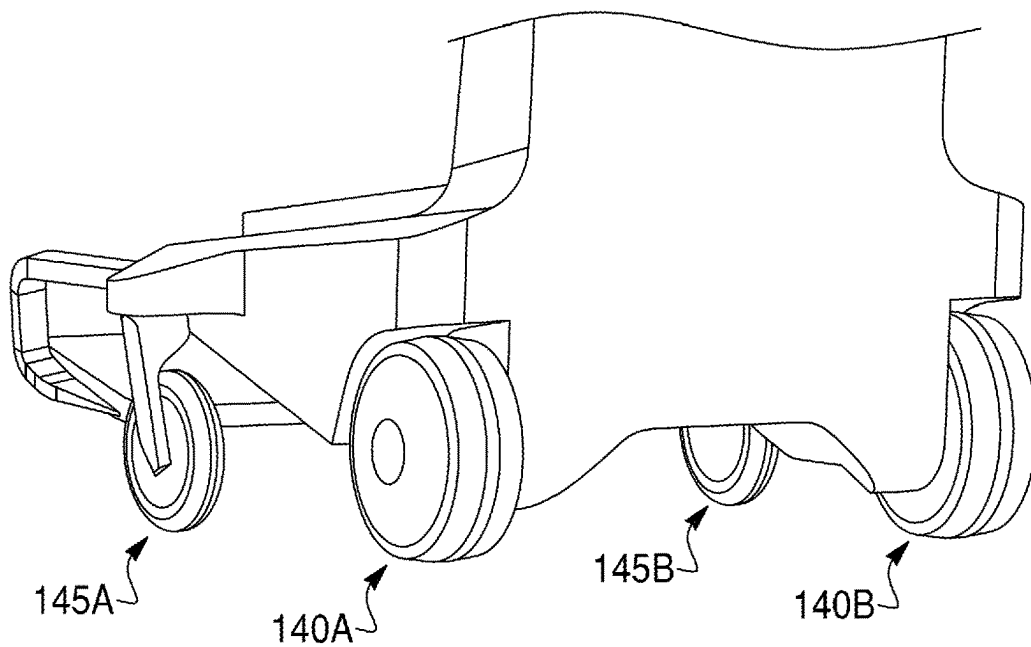
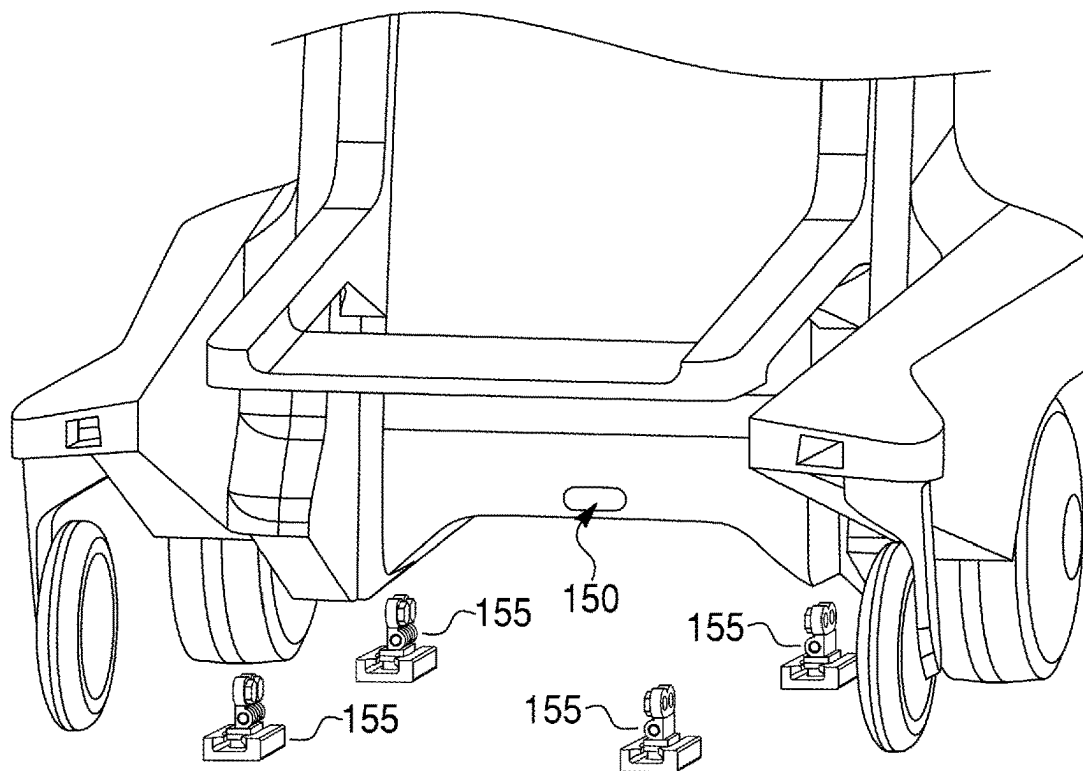


FIG. 7A

**FIG. 8****FIG. 9**

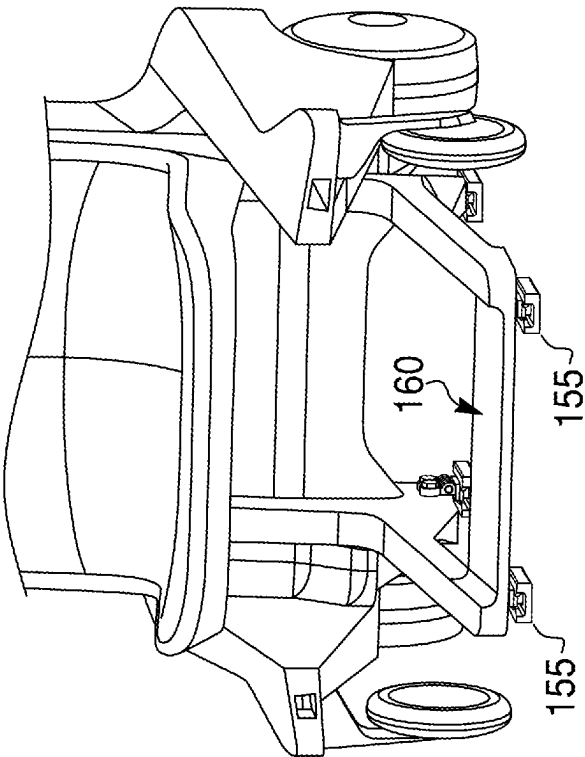


FIG. 10B

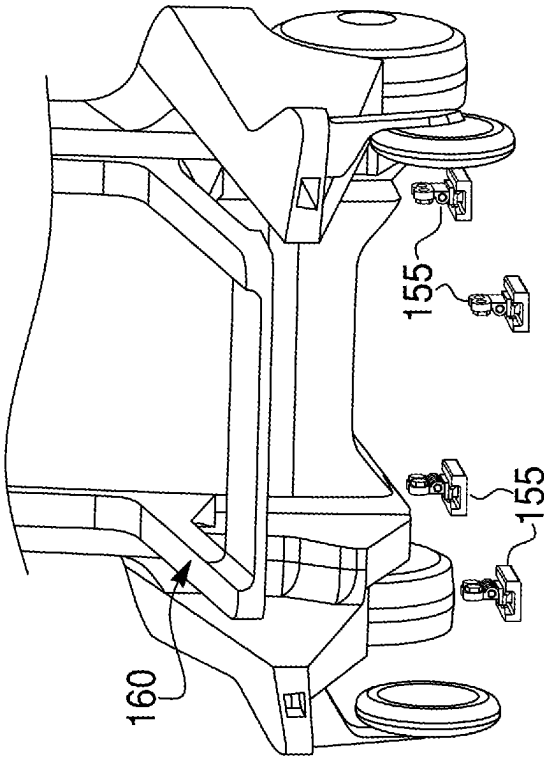


FIG. 10A

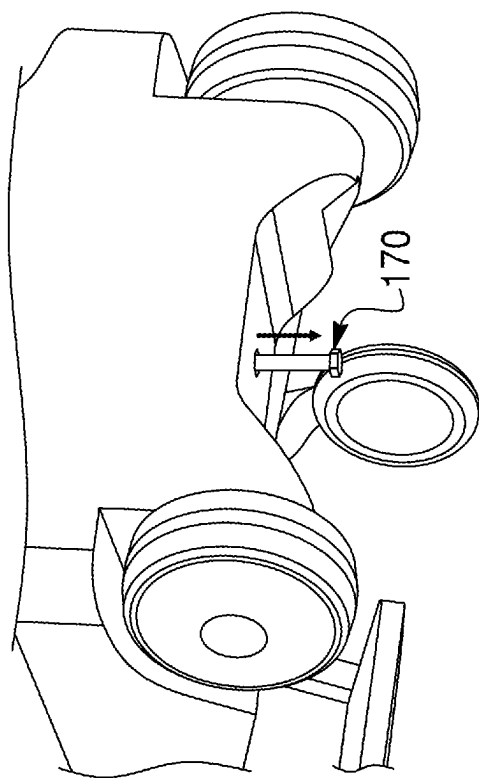


FIG. 11A

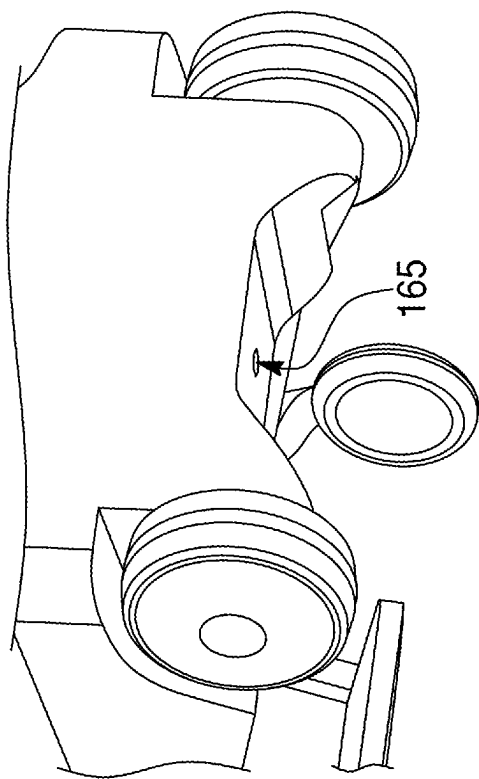


FIG. 11B

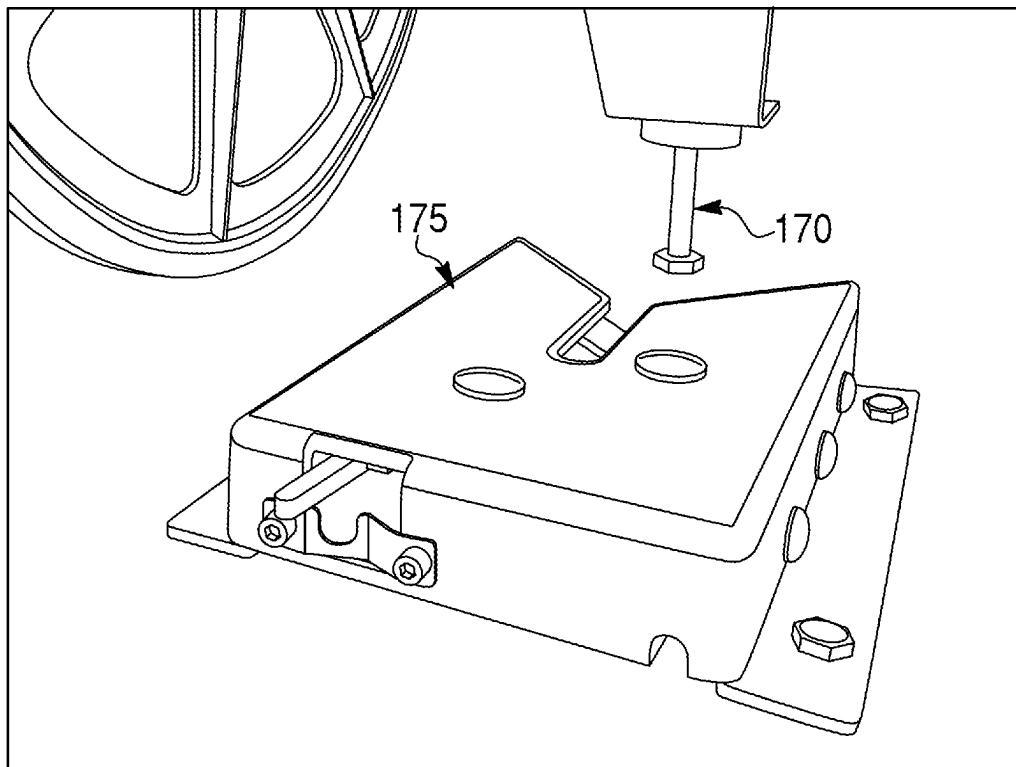


FIG. 12

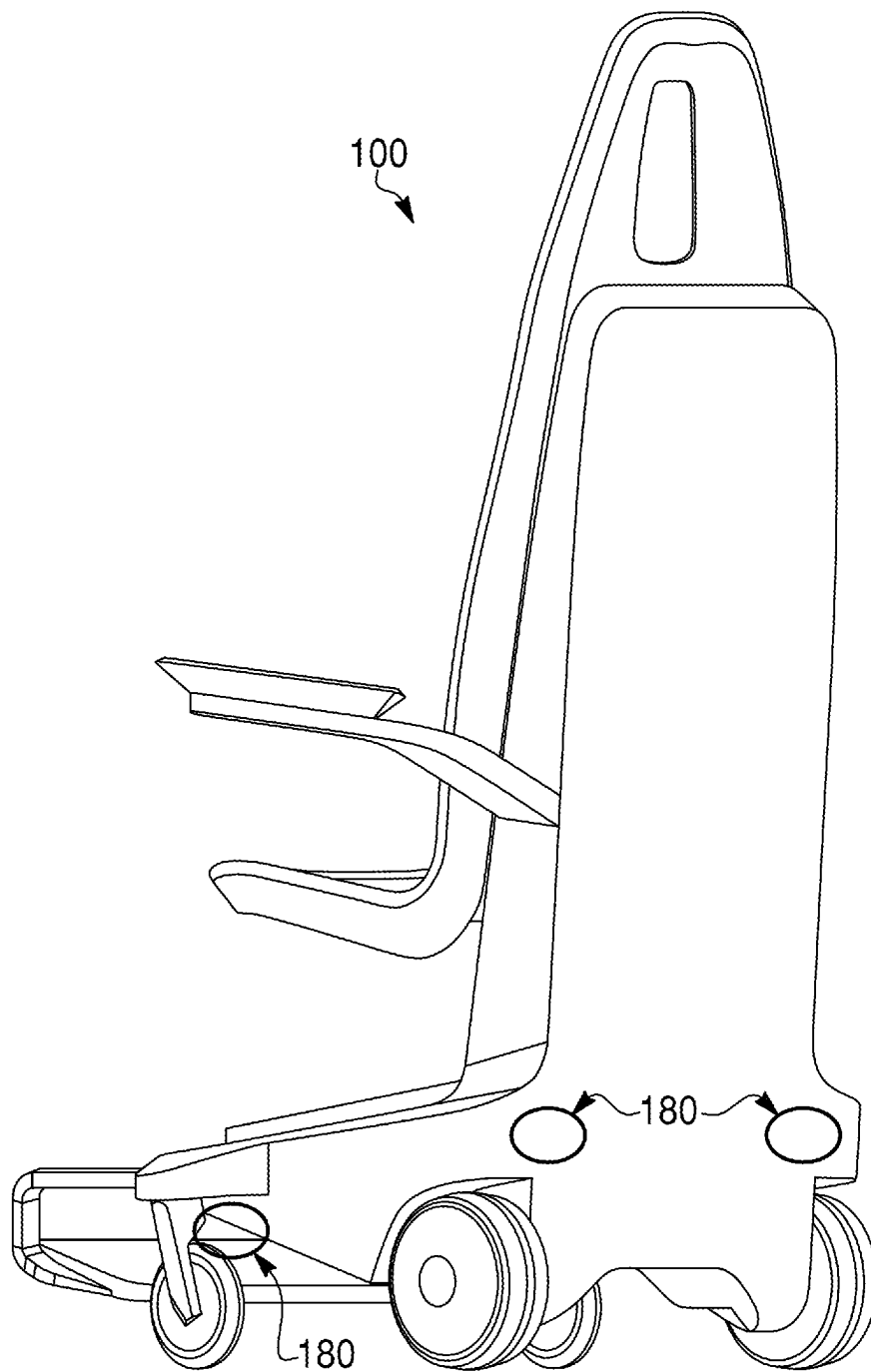


FIG. 13

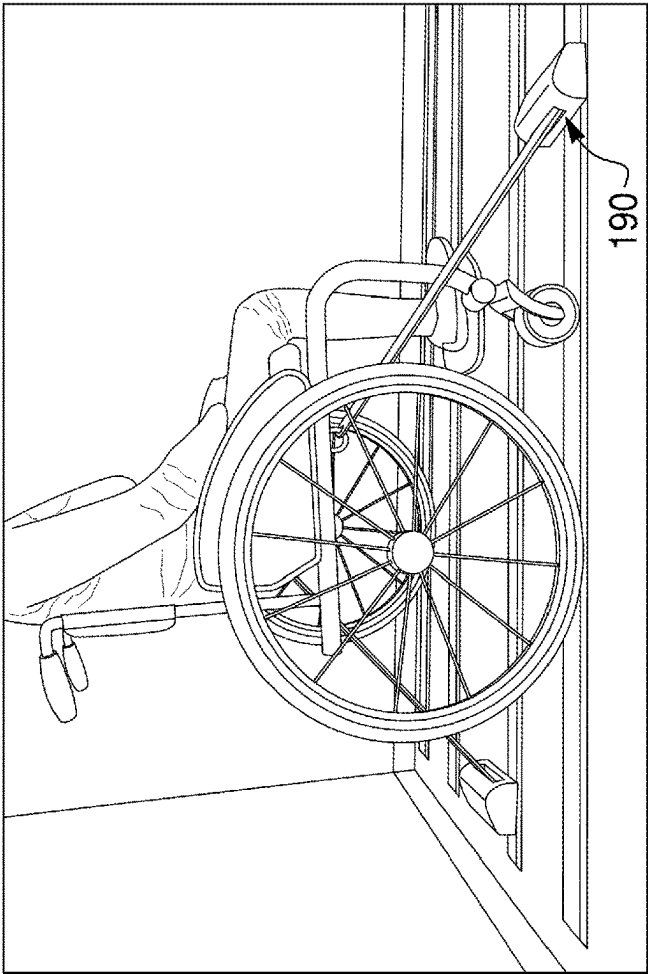


FIG. 14B

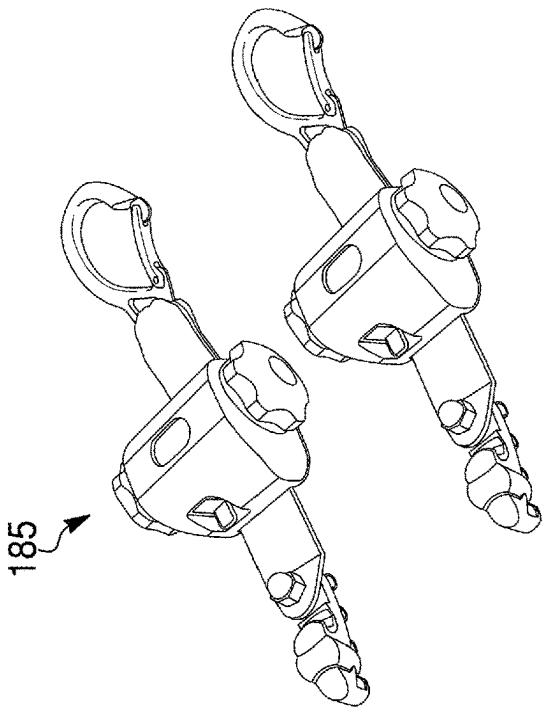


FIG. 14A

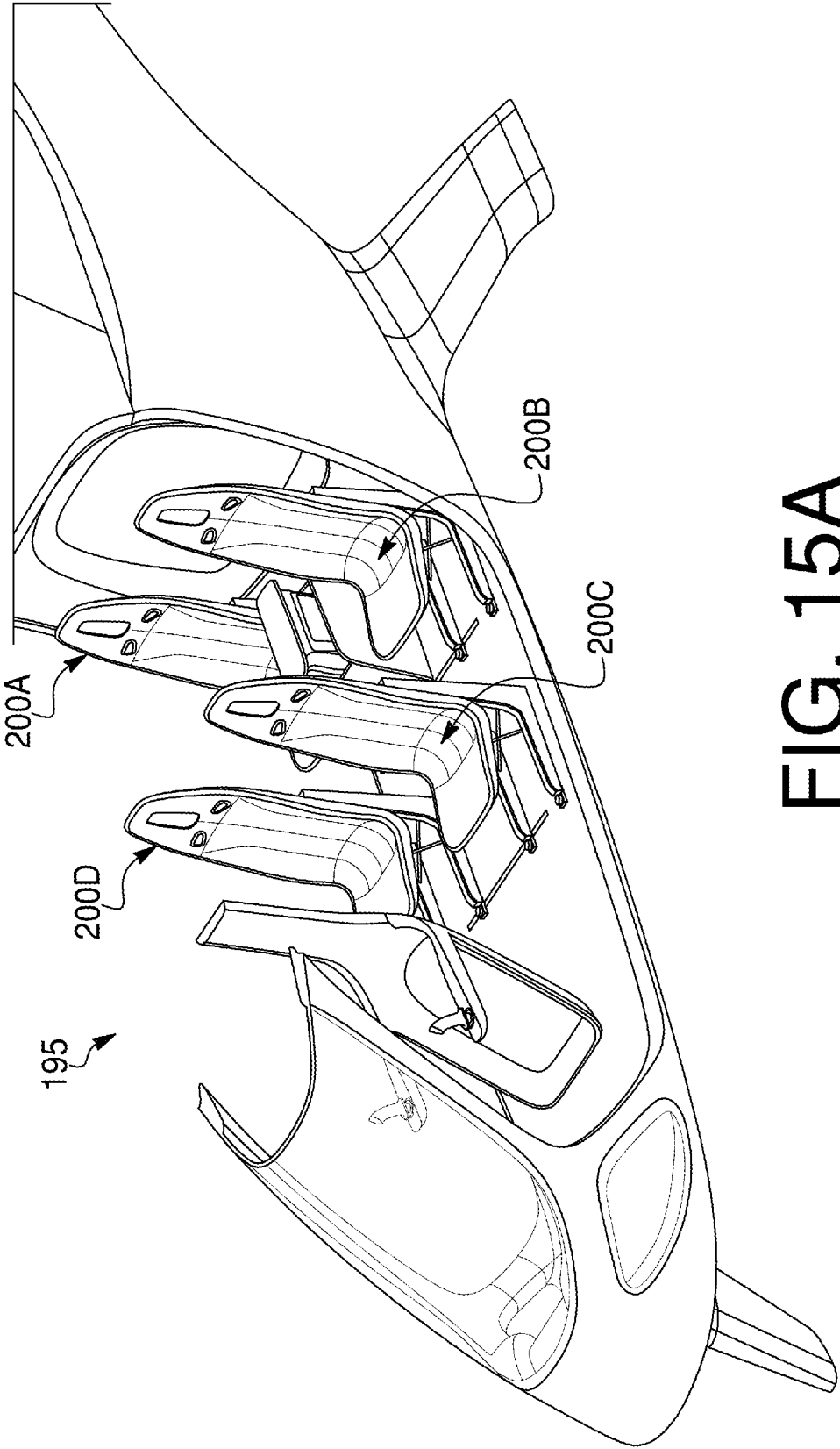


FIG. 15A

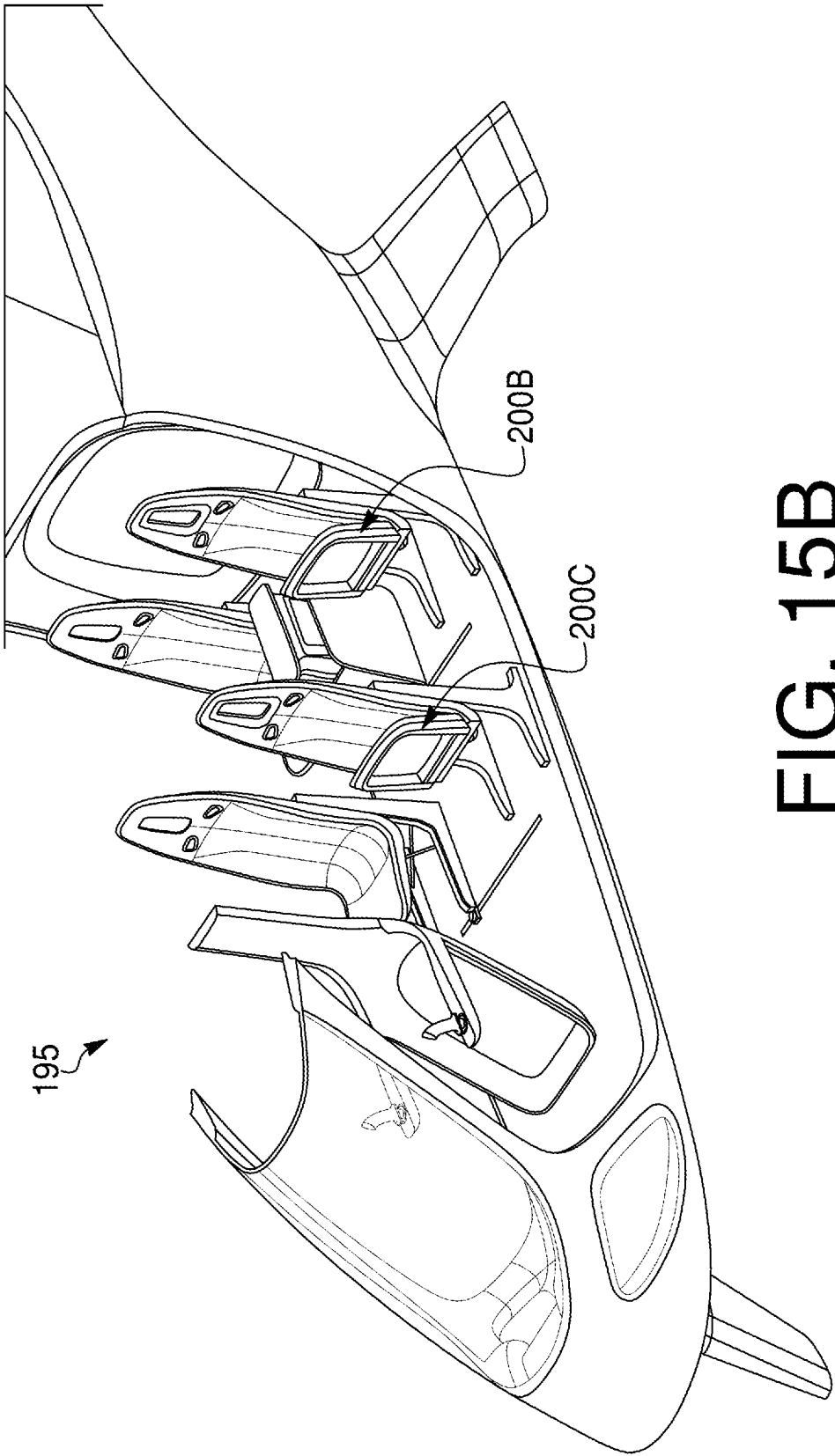


FIG. 15B

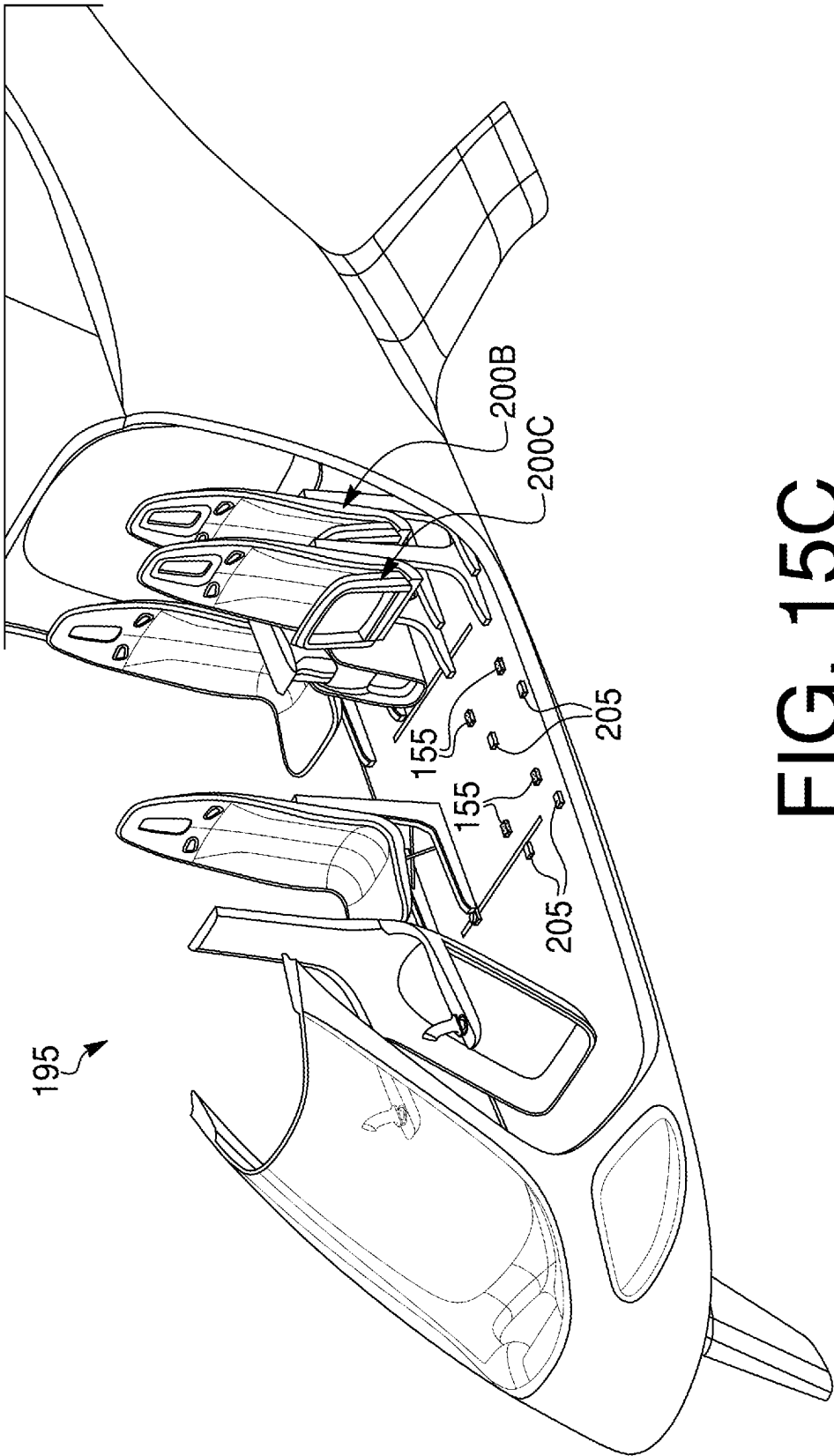


FIG. 15C

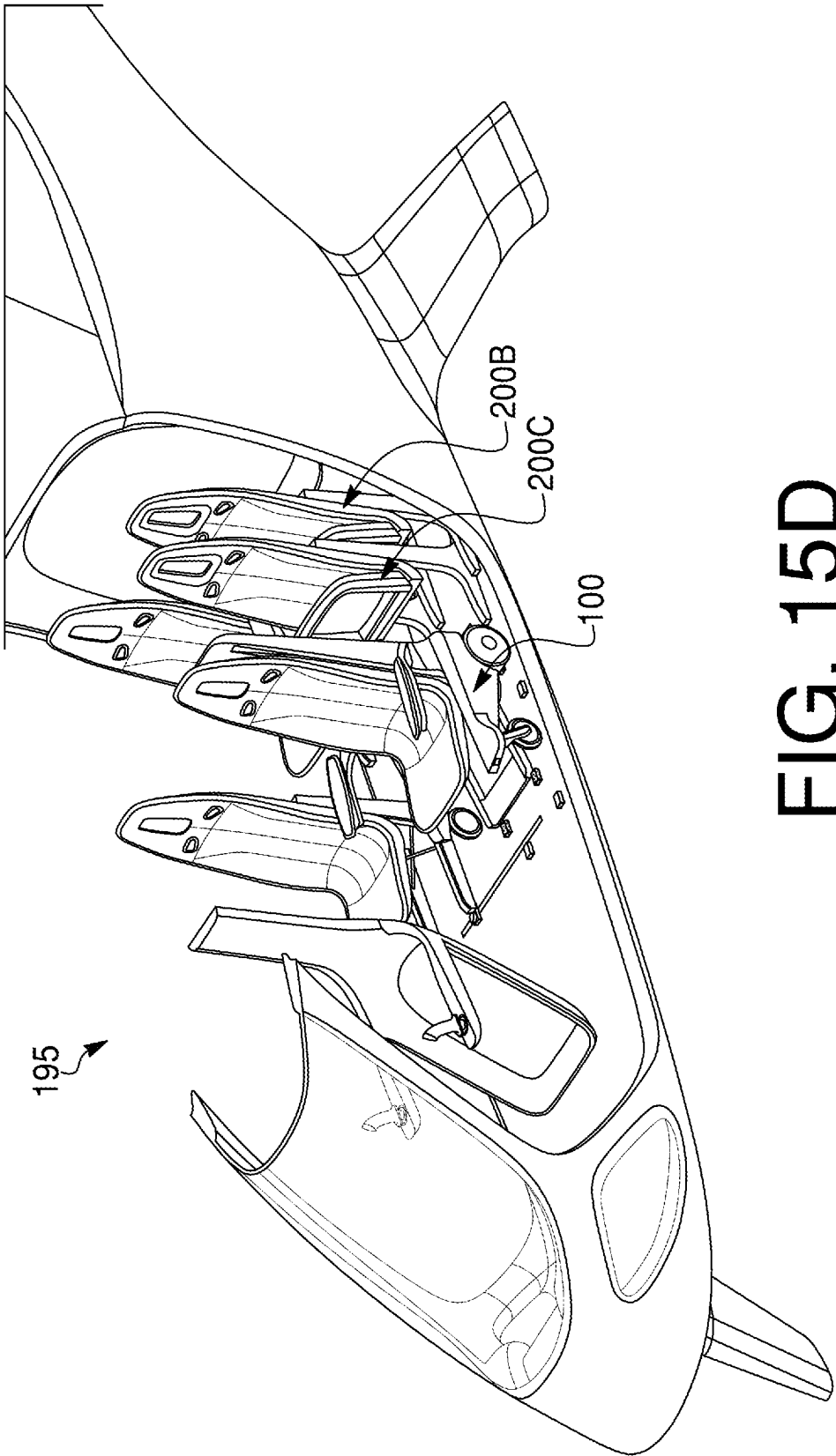


FIG. 15D

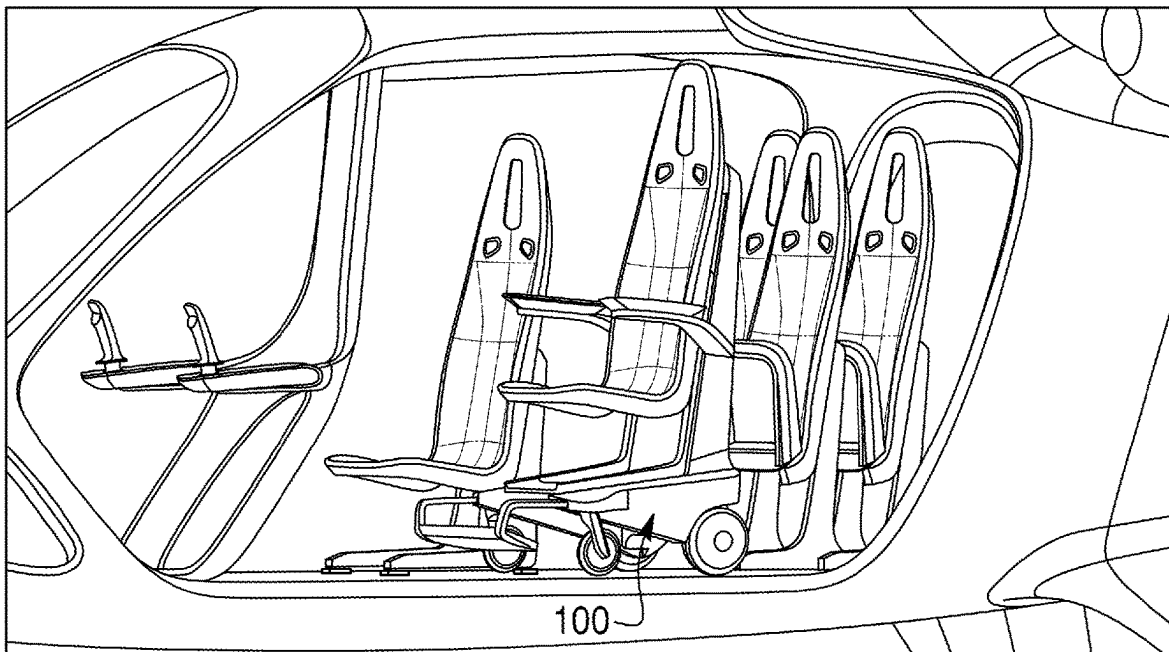


FIG. 16A

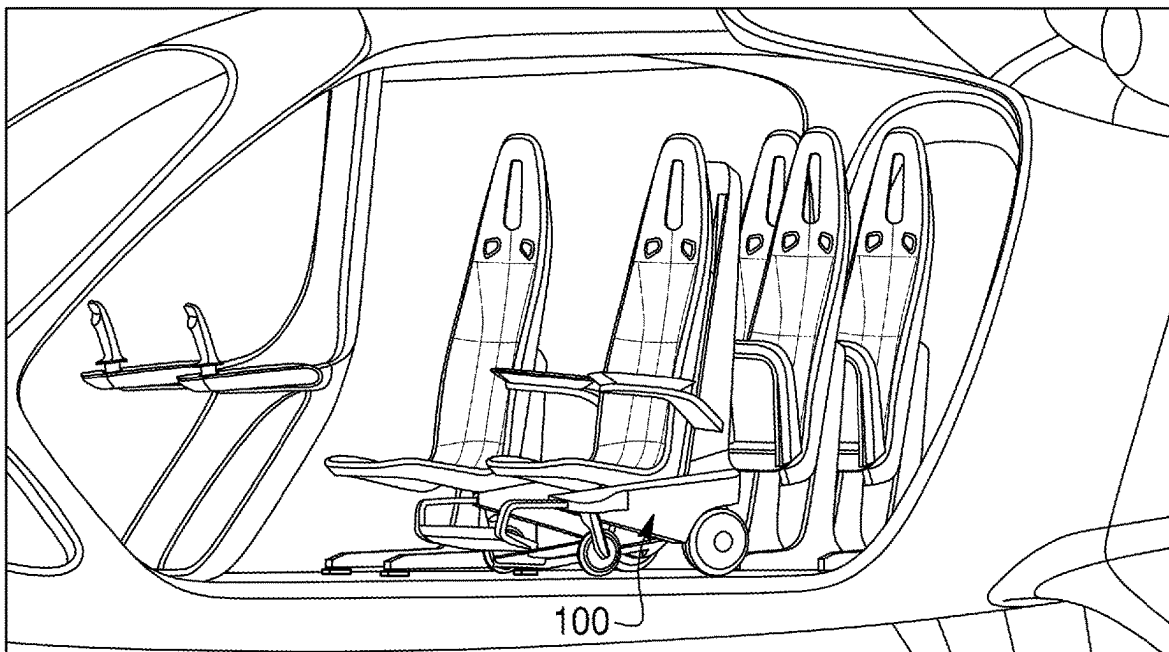
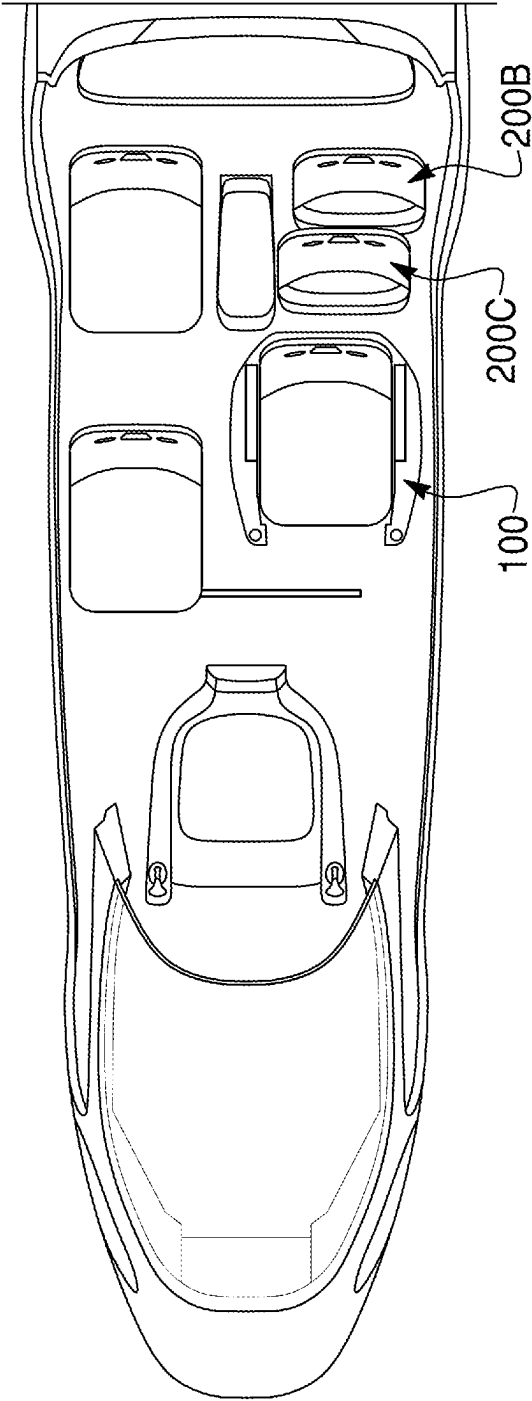


FIG. 16B



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MULTIMODAL WHEELCHAIR AND METHODS THEREOF

CROSS REFERENCE TO RELATED APPLICATIONS

This application claims priority to U.S. Provisional Application No. 63/483,580 filed Feb. 7, 2023, which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

Aspects of the present disclosure relate generally to embodiments of a multimodal wheelchair, and, more specifically, to embodiments of the multimodal wheelchair that facilitate improved operation thereof and that enable secure attachment of the multimodal wheelchair to a vehicle.

BACKGROUND

Differently-abled individuals may encounter challenges when navigating public transportation, including airplanes, trains, buses, etc. While regulatory standards mandate transportation authorities to provide various accommodations to differently-abled individuals, these provisions often fall short in ensuring a dignified travel experience. For instance, individuals relying on wheelchairs may be subject to certain accommodations that may be overly burdensome and/or demeaning. Accordingly, the existing framework, though aimed at inclusivity, highlights the need for more effective and considerate solutions to enhance the travel experience for those with varying functional needs.

The present disclosure is accordingly directed to a multimodal wheelchair that is configured to be attachable to a vehicle and that users may sit in during vehicle operation. The background description provided herein is for the purpose of generally presenting the context of the disclosure. Unless otherwise indicated herein, the materials described in this section are not prior art to the claims in this application and are not admitted to be prior art, or suggestions of the prior art, by inclusion in this section.

SUMMARY OF THE DISCLOSURE

According to certain aspects of the disclosure, a multimodal wheelchair mountable to a vehicle is disclosed.

In one aspect, a multimodal wheelchair is disclosed. The multimodal wheelchair may include: a first section, including: a processing system contained within a housing of the first section; a movement controller component positioned on a portion of the first section that is connected to the processing system; a plurality of wheels that are controllable by the movement controller; and a set of tracks recessed within a first side of the first section; and a second section, including: a seat portion located on a first side of the second section, the seat portion having an integrated harness; and a connection component, coupled to a second side of the second section, that attaches the second section to the first section via coupling with the set of tracks of the first section.

In another aspect, a power chair base of an electric wheelchair is disclosed. The power chair base may include: a first section including a pair of recessed tracks configured to secure a seat section of the electronic wheelchair; a second section, connected to a distal end of the first section, including a plurality of wheels; a processor contained within a housing of the power chair base; and a movement controller positioned on a surface of the first section.

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In yet another aspect, a method of autonomously docking an electronic wheelchair to a vehicle is disclosed. The method may include: receiving, at a system of the electronic wheelchair, sensor data; identifying, from the sensor data and using a processor associated with the system, an indication of a predetermined docking location within the vehicle; aligning, via instructions provided by the processor to a controller of the electronic wheelchair, the electronic wheelchair with the predetermined docking location; and deploying, subsequent to the aligning, a securement mechanism of the electronic wheelchair to attach with a locking mechanism present at the predetermined docking location.

The foregoing is a summary and thus may contain simplifications, generalizations, and omissions of detail; consequently, those skilled in the art will appreciate that the summary is illustrative only and is not intended to be in any way limiting.

For a better understanding of the embodiments, together with other and further features and advantages thereof, reference is made to the following description, taken in conjunction with the accompanying drawings. The scope of the invention will be pointed out in the appended claims.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate the disclosed embodiments, and together with the description, serve to explain the principles of the disclosed embodiments. There are many aspects and embodiments described herein. Those of ordinary skill in the art will readily recognize that the features of a particular aspect or embodiment may be used in conjunction with the features of any or all of the other aspects or embodiments described in this disclosure. In the drawings:

FIG. 1 depicts a perspective view of a multimodal wheelchair, according to various aspects of the present disclosure.

FIGS. 2A-2B depict connection of a seat portion to a base portion of the multimodal wheelchair, according to various aspects of the present disclosure.

FIG. 3 depicts a movement controller of the multimodal wheelchair, according to various aspects of the present disclosure.

FIG. 4 depicts a side view of a multimodal wheelchair, according to various aspects of the present disclosure.

FIGS. 5A-5C depict varying height configurations of a seat portion within the multimodal wheelchair, according to various aspects of the present disclosure.

FIGS. 6A-6B depict varying seating configurations of a user seated in the multimodal wheelchair, according to various aspects of the present disclosure.

FIGS. 7A-7B depict varying views and configurations of the multimodal wheelchair in a crash event, according to various aspects of the present disclosure.

FIG. 8 depicts wheels of the multimodal wheelchair, according to various aspects of the present disclosure.

FIG. 9 depicts an onboard vision system of the multimodal wheelchair, according to various aspects of the present disclosure.

FIGS. 10A-10B depict operation of a securement mechanism of the multimodal wheelchair, according to various aspects of the present disclosure.

FIGS. 11A-11B depict operation of another securement mechanism of the multimodal wheelchair, according to various aspects of the present disclosure.

FIG. 12 depicts a wheelchair securement device, according to various aspects of the present disclosure.

FIG. 13 depict securement mechanism attachment locations on a multimodal wheelchair, according to various aspects of the present disclosure.

FIGS. 14A-14B depict utilization of a securement mechanism to secure a wheelchair to a vehicle, according to various aspects of the present disclosure.

FIGS. 15A-15D depict a process for mounting the multimodal wheelchair to a vehicle, according to various aspects of the present disclosure.

FIGS. 16A-16B depict various height configurations of the wheelchair seat during a docking process, according to various aspects of the present disclosure.

FIG. 17 depicts a top view of the multimodal wheelchair mounted to a vehicle, according to various aspects of the present disclosure.

DETAILED DESCRIPTION OF THE EMBODIMENTS

The terminology used below may be interpreted in its broadest reasonable manner, even though it is being used in conjunction with a detailed description of certain specific examples of the present disclosure. Indeed, certain terms may even be emphasized below; however, any terminology intended to be interpreted in any restricted manner will be overtly and specifically defined as such in this Detailed Description section.

In this this disclosure, the term “based on” means “based at least in part on.” The singular forms “a,” “an,” and “the” include plural referents unless the context dictates otherwise. The term “exemplary” is used in the sense of “example” rather than “ideal.” The terms “comprises,” “comprising,” “includes,” “including,” or other variations thereof, are intended to cover a non-exclusive inclusion such that a process, method, or product that comprises a list of elements does not necessarily include only those elements, but may include other elements not expressly listed or inherent to such a process, method, article, or apparatus. Relative terms, such as “about,” “approximately,” “substantially,” and “generally,” are used to indicate a possible variation of $\pm 10\%$ of a stated or understood value. In addition, the term “between” used in describing ranges of values is intended to include the minimum and maximum values described herein. The use of the term “or” in the claims and specification is used to mean “and/or” unless explicitly indicated to refer to alternatives only or the alternatives are mutually exclusive, although the disclosure supports a definition that refers to only alternatives and “and/or.” As used herein “another” may mean at least a second or more.

As used herein, the term “user” generally encompasses any person that is operating and/or sitting in a mobility aid such as a multimodal wheelchair, as described herein. As used herein, the term “vehicle” may refer to any type of vehicle, e.g., motor vehicles (e.g., cars, trucks, buses, etc.), railed vehicles (e.g., trains, etc.), amphibious vehicles (e.g., boats, etc.), aircraft (e.g., planes, helicopters, etc.), spacecraft, autonomous or semi-autonomous vehicles, and the like. Various embodiments of the present disclosure relate generally to electric vehicles, such as vehicles driven via one or more electric loads, components associated with the electrical loads, and monitoring systems for the electrical loads and/or the components associated with the electrical loads. The electric loads may be in the form of electric motors associated with one or more propellers of a vertical takeoff and landing vehicle.

Efforts have been made to improve the accommodations provided to differently-abled individuals that utilize public transit. However, many conventional attempts to address the challenges that they may face often result in less-than-ideal outcomes. For instance, the current practice of storing wheelchairs during flights and having individuals manually lifted into designated seats may be burdensome and demeaning. Specifically, the foregoing practice may lead to discomfort during the boarding process and may not always be sensitive to an individuals’ needs. The lack of a seamless and dignified solution may be particularly pronounced in smaller aircraft, such as an electric VTOL, which may lack the necessary space to accommodate certain types of wheelchairs (e.g., battery-powered wheelchairs, etc.), thereby potentially restricting travel options that these differently-abled individuals may use. Accordingly, a need exists for an improved wheelchair that may provide the user with expanded functionality during operation and that also may be capable of securely attaching to a vehicle, such as an aircraft.

In view of the issues described above, aspects of this disclosure aim to address the limitations and discomforts associated with conventional solutions by providing an improved wheelchair, or “multimodal” wheelchair, that ensures that differently-abled individuals may experience dignified and functional travel. The multimodal wheelchair may enable users to travels in all forms of transport without exiting their wheelchair. Additionally, characteristics of the wheelchair may be easily adjusted by the user (e.g., chair speed, chair height, chair orientation, etc.) to further meet their travel needs. Furthermore, the multimodal wheelchair may be configured to manually or autonomously dock within certain vehicle cabins to ensure that users are safely secured during travel. In yet a further aspect, the multimodal wheelchair may be configured to allow full crash attenuation to occur.

Reference will now be made in detail to the exemplary embodiments of the present disclosure described below and illustrated in the accompanying drawings. Wherever possible, the same reference numbers will be used throughout the drawings to refer to same or like parts.

Additional objects and advantages of the embodiments will be set forth in part in the description that follows, and in part will be obvious from the description, or may be learned by practice of the embodiments. It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the claims.

Referring now to FIG. 1, an exemplary multimodal wheelchair is illustrated. In an embodiment, multimodal wheelchair 100 may be an electric wheelchair that comprises a first section, e.g., seat portion 105, and a second section, e.g., base portion 110, which are collectively designed to enhance user comfort, functionality, mobility, and overall safety. Seat portion 105 may have a first side, e.g., “front side,” that provides a comfortable and secure seating arrangement for the user. In embodiment, the first side of seat portion 105 may contain integrated harness 115 that may be utilized to effectively secure a user to the seat and ensure stability and safety during vehicle operation. Integrated harness 115 is depicted in FIG. 1 as containing four strap portions that may converge and be connected to a buckle. In other configurations (not illustrated here) integrated harness 115 may contain fewer or greater strap portions and/or may contain two or more buckles. In an embodiment, seat portion 105 may contain a second side,

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e.g., “back side,” that is integrated with connection component **125**, the details of which are further described herein.

In an embodiment, base portion **110** may be battery-powered and may contain an integrated processing system, e.g., situated within housing of base portion **110**. The integrated processing system may interact with one or more controllers and sensors of an integrated sensor system to control functional aspects of multimodal wheelchair **100**. For instance, integrated processing system may enable a user to control: a movement direction of multimodal wheelchair **100**, a movement speed of multimodal wheelchair **100**, a height setting of seat portion **105**, other features of multimodal wheelchair **100**, and the like. In an embodiment, each multimodal wheelchair **100** may be fully customizable to the user's preferences. In this regard, virtually any dimension, feature, and/or aesthetic or technical characteristic of the multimodal wheelchair may be chosen by a user.

In an embodiment, base portion **110** may contain two general sections. The first section may be a columnar section that extends vertically from a second, wheel-supporting section. In an embodiment, the first section may contain a set of tracks **120** that are recessed into one side. The first section may further contain one or more armrests and/or a movement controller. In an embodiment, second section may secure a set of wheels and/or a footrest and may also house one or more deployment mechanisms. Additional details regarding the foregoing features of base portion **110** are further described herein.

Referring now collectively to FIGS. 2A-2B, an attachment process of seat portion **105** to base portion **110** is illustrated. In an embodiment, connection component **125** may be configured to fit into set of tracks **120** of base portion **110**. For instance, FIG. 2A illustrates base portion **110** having two tracks **120** that corresponding vertical elements of connection component **125** may fit and lock into. Once attached, as shown in FIG. 2B, a user may interact with various controls to adjust characteristics of seat portion **105** and/or movement of base portion **110**. More particularly, referring now to FIG. 3, a user may interact with movement controller **130** that comprises joystick **130A** (for providing directional control and/or seat height control), rotatable speed adjustment knob **130B** (for providing speed control), and digital display **130C** (for providing direction indications, speed indications, battery indications, alert notifications, etc.). Movement controller **130** may be mounted to a portion of base portion **110** (e.g., on or near the arm rests, etc.).

It is important to note that the illustrated movement controller **130** in FIG. 3 is only exemplary and is not intended to be limiting. More particularly, the user may interact with another controlling device (e.g., a remote controller, a keypad control, etc.), in lieu of or in addition to movement controller **130**, to control movement and other aspects of the multimodal wheelchair.

In an embodiment, a user may leverage movement controller **130**, or another control feature of the multimodal wheelchair, to implement height adjustments to their seat. For instance, with respect to FIG. 4, a user may provide inputs (e.g., via interaction with movement controller **130**) to a linear actuator (e.g., housed within base portion **110**) to facilitate vertical height adjustments that may move the seat up or down depending on their needs. For example, referring now to FIG. 5A, multimodal wheelchair **100** is presented in a “ride height” configuration, which may be the height of seat portion **105** during normal use (e.g., approximately 8" lift, approximate 20" seat pan). FIG. 5B illustrates a “lock height” configuration, which may be the height of seat

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portion **105** when multimodal wheelchair **100** is locked inside a cabin during vehicle operation (e.g., approximately 0" lift, approximate 12.25" seat pan). FIG. 5C illustrates a “standing height” configuration, which may be the height of seat portion **105** in a situation where a user desires to speak to another individual at eye-level.

In some embodiments, the user may manipulate height of seat portion **105** manually, e.g., via interaction with aspects of movement controller **130** (or another equivalent device). In other embodiments, height of seat portion **105** may be dynamically adjusted, e.g., without receiving any explicit user input. For instance, one or more sensors (e.g., global positioning system (GPS) GPS sensors, location sensors, etc.) may be positioned in seat portion **105**, base portion **110**, both, etc., that may be configured to gather positional/location information associated with multimodal wheelchair **100**. This information may be transmitted to an onboard processor, e.g., positioned in base portion **110**, that may subsequently be configured to control height of seat portion **105** (e.g., by transmitting instructions to a linear actuator) based on the identified location that multimodal wheelchair **100** is determined to be in. For instance, responsive to identifying that multimodal wheelchair **100** has transitioned from a public location (e.g., an airport, etc.) into a vehicle cabin (e.g., airplane cabin), multimodal wheelchair **100** may be configured to dynamically transition from a “ride height” configuration to a “lock height” configuration.

In an embodiment, the adjustments in height of seat portion **105** may cause other aspects of multimodal wheelchair **100** to correspondingly and automatically adjust. For example, referring now to FIGS. 6A-6B, adjustable footrest **135** integrated in second section of base portion **110** is illustrated that may move in concert with the linear actuator (e.g., may expand or retract as the seat height is adjusted). For instance, FIG. 6A illustrates multimodal wheelchair **100** having a seat height adjusted to a normal configuration. During transition of multimodal wheelchair **100** from the normal configuration to a locked configuration (i.e., in which the seat height is lowered), adjustable footrest **135** may correspondingly extend outward to accommodate the user's resulting leg length (e.g., via manual movement by the user, dynamically via a mechanical or electronic connection with the linear actuator, etc.). In an embodiment, adjustable footrest **135** may be adjustable to any position up to an absolute adjustment length or, alternatively, adjustable footrest **135** may have preconfigured adjustment settings, each of which correspond to a specific adjustment length.

In an embodiment, base portion **110** of multimodal wheelchair **100** may be designed to allow full stroke for crash attenuation. For instance, FIG. 7A depicts the movement range of seat portion **105** in multimodal wheelchair **100** during a crash event. Specifically, the linear actuator may be configured to release seat portion **105** from its present height configuration with respect to connection component **125** to a lowered configuration (e.g., in which seat portion **105** contacts protrusions **125a** of connection component **125**) in response to detection of a crash event, thereby decreasing the forces that may act on a user's body during a crash event. FIG. 7B depicts a side view of the movement range of seat portion **105** occurring in FIG. 7A.

Referring now to FIG. 8, base portion **110** of the multimodal wheelchair **100** may contain a plurality of wheels. More particularly, base portion **110** may contain four wheels via which multimodal wheelchair **100** may move around on. Each of the four wheels may be composed of a non-marking rubber material. In an embodiment, the two rear wheels **140A, B** may be 8" independent power wheels with electric

HUB motors whereas the two front wheels **145A, B** may be 5" caster wheels. It is important to note that these wheel dimensions are not limiting and other wheelchair configurations having different sized wheels may also be possible. For instance, in an embodiment, multimodal wheelchair **100** may contain fewer or greater than four wheels. In another embodiment, each of the wheels of multimodal wheelchair **100** may be substantially the same size and/or each of the wheels may be independent power wheels having independent electric HUB motors.

Referring now to FIG. **9**, multimodal wheelchair **100** may contain an integrated sensor system, also described herein as an "onboard vision system," that may contain one or more sensors (e.g., light/camera sensors, proximity sensors, infrared sensors, etc.) that are capable of detecting objects positioned around multimodal wheelchair **100**. For instance, FIG. **9** illustrates a sensor **150** associated with the onboard vision system that is capable of identifying objects, and/or the characteristics thereof, within a viewing/detection range of sensor. In an embodiment, data obtained by sensor **150** may be leveraged by the processing system to facilitate autonomous docking with a vehicle, e.g., an aircraft. More particularly, the onboard vision system may leverage one or more sensors to identify a predetermined location within the cabin where multimodal wheelchair **100** must be situated. For instance, the vehicle cabin may contain four locking mechanisms **155** that are positioned at a predetermined location on the cabin floor. The sensor(s) of onboard vision system may detect the positions of these locking mechanisms **155** (e.g., via image analysis and object recognition, etc.) and transmit indications of their identified positions to a system processor. The system processor may thereafter dynamically steer multimodal wheelchair **100** (e.g., by providing instructions to one or more controllers) so that it is properly aligned with locking mechanisms **155**. In some aspects, locking mechanisms **155** may be stored within the vehicle floor and may deploy, e.g., either by manual actuation or by dynamic deployment (e.g., upon receiving an indication from a processor/controller that a multimodal wheelchair **100** is proximate to a position of the locking mechanisms), to accommodate multimodal wheelchair **100**.

Referring now to FIGS. **10A** and **10B**, once multimodal wheelchair **100** is properly positioned with respect to locking mechanisms **155**, the system processor may provide instructions (e.g., to one or more relevant controllers contained in base portion **110**, etc.) to deploy securement mechanism **160**, which may effectively secure multimodal wheelchair **100** in place. More particularly, FIG. **10B** illustrates that securement mechanism **160** may be lowered down to initiate coupling with locking mechanisms **155**. Upon proper attachment, multimodal wheelchair **100** may be securely held in place during the duration of vehicle operation.

It is important to note that the number, position, and/or type of locking mechanisms are not limited to what is presented in FIGS. **10A** and **10B**. For example, multimodal wheelchair **100** may be configured to attach to fewer or greater locking mechanisms. Similarly, it is also important to note that the number, shape, and/or dimensions of securement mechanism **160** is not limited to the implementation illustrated in FIGS. **10A** and **10B**.

In an embodiment, the wheelchair user and/or one or more other individuals may be apprised when a successful docking is occurring. For instance, a notification indicating that multimodal wheelchair **100** successfully docked with locking mechanisms **155** may be transmitted to digital display **130C** of movement controller **130**, to a user's personal

device (e.g., a user's mobile device, etc.), a pilot's personal device, any combination of the foregoing, and the like. Similarly, a notification may be transmitted to any of the foregoing devices responsive to detecting that multimodal wheelchair **100** becomes unsecured during flight. In an embodiment, certain vehicle functions may be limited unless an indication of successful docking of the multimodal wheelchair is received. For instance, a pilot may not be able to activate various aircraft systems that facilitate flight unless the aircraft systems are apprised that the multimodal wheelchair is secured to the aircraft.

FIGS. **11-14B** illustrate different types of securement mechanisms and techniques for securing multimodal wheelchair **100** within a vehicle during travel.

Referring now to FIGS. **11A** and **11B**, a deployable securement mechanism for attaching the multimodal wheelchair to a vehicle (e.g., car, train, bus, boat, etc.) is provided. In FIG. **11A**, multimodal wheelchair **100** is illustrated as having deployment port **165** (e.g., on the underside of base portion **110**). Securement mechanism **170** may be stowed within the housing of base portion **110** when not in use and may deploy through deployment port **165** when released, as illustrated in FIG. **11B**. In an embodiment, securement mechanism **170** may be deployed manually (e.g., by the user interacting with movement controller **130** or another device) or, alternatively, may be deployed dynamically (e.g., automatically upon the sensor system automatically identifying that multimodal wheelchair **100** is within a predetermined distance of a detected locking mechanism). In an embodiment, the securement mechanism illustrated here may be a bolt (e.g., a Q'STRAIN bolt, etc.) that may interact with a corresponding locking mechanism that is integrated into the vehicle floor, such as locking mechanism **175** illustrated in FIG. **12**. It is important to note that the securement mechanism illustrated in FIGS. **11B** and **12** is not limiting and another securement mechanism having a different shape, size, position, and/or be of a different type may be utilized.

Referring now to FIG. **13**, multimodal wheelchair **100** is illustrated that contains a plurality of securement mechanism attachment locations **180**. One or more tie down securement mechanisms **185**, such as those illustrated in FIG. **14A**, may contain a first end and a second end. The first end of tie down securement mechanisms **185** may be attached to the securement mechanism attachment locations **180** of the multimodal wheelchair **100** and a second end may be attached to a portion of track **190** that is integrated into a vehicle floor, as illustrated in FIG. **14B**. The securement mechanism described in FIGS. **13** and **14** may be utilized in addition to or in the absence of the securement mechanism described in FIGS. **11** and **12**.

Referring now to FIGS. **15A** and **15D**, a wheelchair docking process for securing a multimodal wheelchair within a passenger cabin of an aircraft is illustrated. FIG. **15A** illustrates cabin **195** of a small aircraft, such as an EVTOL vehicle. In normal operation, cabin **195** may contain four passenger seats **200** (A-D). In an embodiment, one or more of these seats may be adjusted and moved to accommodate an individual using a multimodal wheelchair. For instance, FIGS. **15B** and **15C** illustrate that the two seats **200B, C** may be folded up and moved to a rear portion of cabin **195**. In this regard, seat **200C** may be removed from its original position (represented by securement holes **205**) and moved back toward another set of securement holes (not illustrated) near seat **200B**. Thereafter, multimodal wheelchair **100** may align with locking mechanisms **155** and commence docking, as illustrated in FIG. **15D** and as previously described above. FIGS. **16A** and **16B** provide

additional views of multimodal wheelchair **100** during the docking process. More particularly, FIG. **16A** illustrates multimodal wheelchair **100** in a “ride height” configuration in which it has aligned with locking mechanisms **155**. FIG. **16B** illustrates a transition of seat portion **105** of multimodal wheelchair **100** from the ride height configuration to a lock height configuration, in which securement mechanism **160** engages locking mechanisms **155**, thereby securing multimodal wheelchair **100** in place. FIG. **17** illustrates a top-level view of a docked multimodal wheelchair **100** in an aircraft cabin subsequent to the docking processes described in FIGS. **15** and **16**.

The many features and advantages of the present disclosure are apparent from the detailed specification, and thus, it is intended by the appended claims to cover all such features and advantages of the present disclosure that fall within the true spirit and scope of the disclosure. Further, since numerous modifications and variations will readily occur to those skilled in the art, it is not desired to limit the present disclosure to the exact construction and operation illustrated and described, and accordingly, all suitable modifications and equivalents may be resorted to, falling within the scope of the present disclosure.

Moreover, those skilled in the art will appreciate that the conception upon which this disclosure is based may readily be used as a basis for designing other structures, methods, and systems for carrying out the several purposes of the present disclosure. Accordingly, the claims are not to be considered as limited by the foregoing description.

What is claimed is:

1. A multimodal wheelchair, comprising:
 - a first section, including:
 - a processing system contained within a housing of the first section;
 - a movement controller component positioned on a portion of the first section that is connected to the processing system;
 - a plurality of wheels that are controllable by the movement controller; and
 - a set of tracks recessed within a first side of the first section; and
 - a second section, including:
 - a seat portion located on a first side of the second section, the seat portion having an integrated harness; and
 - a connection component, coupled to a second side of the second section, that attaches the second section to the first section via coupling with the set of tracks of the first section.
2. The multimodal wheelchair of claim **1**, wherein the movement controller component comprises: a joystick controller, a speed adjustment knob, and a digital display.
3. The multimodal wheelchair of claim **1**, wherein a height of the second section is adjustable with respect to the first section via user interaction with the movement controller portion.
4. The multimodal wheelchair of claim **1**, wherein the seat portion is configured to vertically move relative to the connection component in response to a crash event.
5. The multimodal wheelchair of claim **1**, wherein the first section further includes a securement mechanism configured to deploy from within the housing of the first section through a deployment port.
6. The multimodal wheelchair of claim **1**, wherein the first section further comprises a footrest that is configured to dynamically expand or retract based on vertical movement of the second section with respect to the first section.

7. The multimodal wheelchair of claim **1**, further comprising at least one armrest integrated with the first section.

8. The multimodal wheelchair of claim **7**, wherein the movement controller is positioned on the at least one armrest.

9. The multimodal wheelchair of claim **1**, further comprising a sensor system, operably coupled to the processing system, containing at least one sensor, wherein the sensor system is configured to enable autonomous docking of the multimodal wheelchair with a locking mechanism of a vehicle.

10. The multimodal wheelchair of claim **9**, wherein the at least one sensor comprises at least one of: a camera sensor, a proximity sensor, and an infrared sensor.

11. A power chair base of an electronic wheelchair, comprising:

- a first section including a pair of recessed tracks configured to secure a seat section of the electronic wheelchair;
- a second section, connected to a distal end of the first section, including a plurality of wheels;
- a processor contained within a housing of the power chair base; and
- a movement controller positioned on a surface of the first section.

12. The power chair base of claim **11**, further comprising a sensor system containing at least one sensor that is operably coupled to the processor.

13. The power chair base of claim **11**, wherein manipulation of the movement controller is configured to cause the processor to adjust one or more of: a speed of the power chair base, a movement direction of the power chair base, and a height of the seat section.

14. The power chair base of claim **11**, further comprising a deployable securement mechanism situated within the housing.

15. A method of autonomously docking an electronic wheelchair to a vehicle, the method comprising:

- receiving, at a system of the electronic wheelchair, sensor data;
- identifying, from the sensor data and using a processor associated with the system, an indication of a predetermined docking location within the vehicle;
- aligning, via instructions provided by the processor to a controller of the electronic wheelchair, the electronic wheelchair with the predetermined docking location; and
- deploying, subsequent to the aligning, a securement mechanism of the electronic wheelchair to attach with a locking mechanism present at the predetermined docking location.

16. The method of claim **15**, wherein the indication of the predetermined docking location corresponds to identification of at least one of: a predefined object or a predefined marking.

17. The method of claim **15**, wherein the securement mechanism is a bolt configured to deploy through a deployment hole from within the electronic wheelchair.

18. The method of claim **15**, wherein the securement mechanism is a height-adjustable portion of the electronic wheelchair that is shaped to interact with connection points of the locking mechanism.

19. The method of claim **15**, further comprising transmitting a docking notification to a designated device.

20. The method of claim **19**, wherein the designated device is at least one of: a personal computing device

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associated with a user, a movement controller of the electronic wheelchair, and another device associated with a pilot of the vehicle.

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